

Yu Gu

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Research interests: agent safety and interpretability; reliable LLM reasoning; uncertainty quantification; knowledge editing; efficient adaptation

EDUCATION

PhD in Mathematics and Statistics , McGill University, Montreal, QC, Canada GPA: 4.00/4.00	Sep 2022- Present
MA in Statistics , Columbia University, New York, NY, USA GPA: 3.95/4.00	Sep 2020- Feb 2022
BMath (Honours) , University of Waterloo, Waterloo, ON, Canada GPA: 3.60/4.00	Sep 2016- Jun 2020

RESEARCH EXPERIENCE

Conformal Aggregation for Chain-of-Thought Reasoning , PhD Research Project • Developed an inference-time reliability framework for multi-path LLM reasoning using quality-weighted aggregation and conformal calibration of abstention. • Proved finite-sample control of confident error and characterized score separability as the condition under which abstention improves selective accuracy. • Evaluated 10 model-dataset settings across GSM8K, MATH, MATH-Hard, and HotpotQA; achieved 90.1% selective accuracy on GSM8K while abstaining on fewer than 5% of questions, versus 82% under majority voting.	2026-Present
Deep Neural Networks for Survival Data Prediction , PhD Research Project <i>Advisor: Prof. Masoud Asgharian</i> • Designed deep-learning models and developed theoretical results for prediction and interpretation with censored survival data.	May 2023- Present
Experimental Data Analysis of Random Geometry and Topology , Columbia University <i>Advisor: Prof. Andrew Blumberg</i> • Implemented Smith normal form and stochastic random-geometry simulations in Python and analyzed the resulting geometric features.	Jan-Apr 2021
Statistical Analysis of Oncogene Effects on Tumor Cells , Nanjing Medical University <i>Advisor: Prof. Li Yuan</i> • Analyzed assay and clinical data using hypothesis tests, Kaplan-Meier estimation, and Cox proportional-hazards models.	May 2018- Jan 2019

PROFESSIONAL AND TEACHING EXPERIENCE

Associate AI Researcher , Quantum Technology Inc., Quebec, Canada • Reviewed and developed LLM post-training, compression, and quantization methods and explored efficient architectures for faster inference.	Jun 2023-Jun 2024 Apr-Nov 2025
Teaching Assistant , Fundamentals of Statistical Learning, McGill University • Led weekly tutorials on programming and machine-learning theory, revised notes and code examples, and graded coursework.	Jan-May 2023

TECHNICAL SKILLS

Programming and data analysis: Python, MATLAB, R, SPSS, Stata

HONORS AND AWARDS

GERAD Scholarship for PhD Students (CAD 10,000), McGill University	Fall 2023
Chair's List of Academic Achievement, Columbia University	Spring 2021
Term Dean's Honors List, University of Waterloo	Winter 2019
President's Scholarship, University of Waterloo	2016
Second Prize, Euclid Mathematics Contest (top 100 of 16,180)	2016
Second Prize, Chinese National High School Mathematics Contest (top 1,000 of 100,000)	2014

Publications and Preprints

Publication record updated September 29, 2026 | * Equal contribution

PEER-REVIEWED PUBLICATIONS

1. **Yu Gu***, Zijun Yu*, Vahid Partovi Nia, and Masoud Asgharian. "Pause and Reflect: Conformal Aggregation for Chain-of-Thought Reasoning." *Advances in Neural Information Processing Systems (NeurIPS 2026)*, accepted. [arXiv:2605.14098](#)
2. **Yu Gu***, Zijun Yu*, Chi-Kuang Yeh, Xinyu Wang, and Ziyang Song. "Severity-Controlled Prediction Sets for Medication Recommendation." *Advances in Neural Information Processing Systems (NeurIPS 2026)*, accepted (poster).
3. Xinyu Wang*, Sicheng Lyu*, **Yu Gu***, Jerry Huang, Peng Lu, Yufei Cui, and Xiao-Wen Chang. "Beyond Hard Writes and Rigid Preservation: Soft Recursive Least-Squares for Lifelong LLM Editing." *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026)*, to appear. [arXiv:2601.15686](#)
4. Xinyu Wang, Ziyu Zhao, Peng Lu, **Yu Gu**, and Xiao-Wen Chang. "OBJKQ: Objective-Joint Babai-Klein Quantization." *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026)*, to appear. [arXiv:2602.08376](#)
5. Sicheng Lyu, **Yu Gu**, Xinyu Wang, Jerry Huang, Sitao Luan, Yufei Cui, Xiao-Wen Chang, and Peng Lu. "EvoEdit: Evolving Null-space Alignment for Robust and Efficient Knowledge Editing." *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 1520-1540, 2026. [doi:10.18653/v1/2026.findings-acl.75](#)
6. Zijun Yu, **Yu Gu**, Vahid Partovi Nia, and Masoud Asgharian. "G-CARB: Graph-Localized Conformal Agent Risk Budget for Compositional Harm." *NeurIPS 2026 Workshop (SLM-Agents)*, accepted.
7. Zhenyao Chen, Xin Chen, Tianyao Lei, **Yu Gu**, Jinyao Gu, Jiali Huang, Binbin Lu, Li Yuan, Ming Sun, and Zhaoxia Wang. "Integrative Analysis of NSCLC Identifies LINC01234 as an Oncogenic lncRNA that Interacts with HNRNPA2B1 and Regulates miR-106b Biogenesis." *Molecular Therapy*, 28(6):1479-1493, 2020. [doi:10.1016/j.ymthe.2020.03.010](#)
8. Zhenyao Chen, Xin Chen, Binbin Lu, **Yu Gu**, Qinnan Chen, Tianyao Lei, Fengqi Nie, Jinyao Gu, Jiali Huang, Chenchen Wei, Ming Sun, and Zhaoxia Wang. "Up-regulated LINC01234 Promotes Non-small-cell Lung Cancer Cell Metastasis by Activating VAV3 and Repressing BTG2 Expression." *Journal of Hematology & Oncology*, 13:7, 2020. [doi:10.1186/s13045-019-0842-2](#)

PREPRINTS

9. Mohammadreza Sadeghi, Mahsa Ghazvini Nejad, MirHamed Jafarzadeh Asl, **Yu Gu**, Yuanhao Yu, Masoud Asgharian, and Vahid Partovi Nia. "MoKA: Mixture of Kronecker Adapters." *arXiv:2508.03527*, 2025. [arXiv link](#)